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Section : SE-1B

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Class : Programming lab

Task 1

```
#include <stdio.h>

int main() {
 int year;

 printf("Enter year: ");
 scanf("%d", &year);

 if (year % 4 == 0) {

 if (year % 100 == 0) {

 if (year % 400 == 0) {
 printf("Leap Year");
 }
 else {
 printf("Not a Leap Year");
 }

 }
 else {
 printf("Leap Year");
 }

 }
```

```
 else {
 printf("Not a Leap Year");
 }
 return 0;
}
```

Task2 #include

```
<stdio.h>

int main() {
 int a, b, c;

 printf("Enter three sides: ");
scanf("%d %d %d", &a, &b,
&c);

 if (a < b + c) {

 if (b < a + c) {

 if (c < a + b) {

 if (a == b) {

 if (b == c) {
 printf("Equilateral");  }
 else {
 printf("Isosceles");  }

 }
 else {

 if (b == c) {
 printf("Isosceles");  }
 else {

 if (a == c) {
 printf("Isosceles");  }
```

```
else {
printf("Scalene");
}

}

}

}
else {
printf("Not a valid triangle");
}

}
else {
printf("Not a valid triangle");
}

}
else {
printf("Not a valid triangle");
}

return 0;
}
```

Task 3

```
#include <stdio.h>

int main() {
 int x, y;

 printf("Enter x and y coordinates: ");
 scanf("%d %d", &x, &y);

 if (x == 0) {

 if (y == 0) {
 printf("Origin");
 }
```

```
 else {
 printf("On the Y-axis");  }

 }
 else if (y == 0) {
 printf("On the X-axis");  }
 else if (x > 0) {

 if (y > 0) {
 printf("Quadrant I");
 }
 else {
 printf("Quadrant IV");  }

 }
 else {

 if (y > 0) {
 printf("Quadrant II");  }
 else {
 printf("Quadrant III");  }

 }

 return 0;
}
```

Task 4

```
#include <stdio.h>

int main() {
 float weight, height, bmi;

 printf("Enter weight in kg: ");
scanf("%f", &weight);
 printf("Enter height in metres: ");
scanf("%f", &height);

 bmi = weight / (height * height);
 printf("BMI = %.2f\n", bmi);

 if (bmi < 18.5) {
 printf("Category: Underweight");
```

```
}
else if (bmi < 25) {
printf("Category: Normal");
}
else if (bmi < 30) {
printf("Category: Overweight");
}
else {
printf("Category: Obese");
}

return 0;
}
```

Task 5

```
#include <stdio.h>

int main() {
 int a, b, c;

 printf("Enter three numbers: ");
 scanf("%d %d %d", &a, &b, &c);

 if (a > b) {

 if (a > c) {
 printf("a is the largest\n");
 printf("Largest value = %d", a);
 }
 else {
 printf("c is the largest\n");
 printf("Largest value = %d", c);
 }

 }
 else {

 if (b > c) {
 printf("b is the largest\n");
```

```
printf("Largest value = %d", b);  }
 else {
 printf("c is the largest\n");
printf("Largest value = %d", c);  }

 }

 return 0;
}
```

Task 6

```
#include <stdio.h>

int main() {
 int w;

 printf("Enter watermelon weight: ");
scanf("%d", &w);

 if (w % 2 == 0) {

 if (w >= 4) {
 printf("YES");
 }
 else {
 printf("NO");
 }

 }
 else {
 printf("NO");
 }
 return 0;
}
```